

Spiking Neural Network (SNN) Robustness Benchmark
ArcTangent (α) stiffness sweeps under Data Heterogeneity and Attacks

ByzFL SNN Benchmark

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1 Introduction and Experimental Settings

This report documents the robustness sweeps of Spiking Neural Networks (SNN) using the ArcTangent (`atan`) surrogate gradient. The stiffness parameter α is swept across 7 values: $\alpha \in \{0.5, 0.75, 1.0, 1.2, 1.4, 1.5, 2.0\}$.

The performance of the model is evaluated in federated learning under varying data heterogeneity (non-IID) levels, under Byzantine attacks, using different robust aggregators. For each configuration, the Spiking Neural Network results are compared side-by-side with a standard (non-spiking) Convolutional Neural Network baseline (`cnn_mnist`) evaluated under the same conditions.

1.1 Parameter Configurations

Table 1 details the parameters, hyperparameters, and algorithms used in this sweep.

Table 1: Experiment Parameters and Hyperparameters

Component / Parameter	Configuration Value
Dataset	MNIST (60,000 train, 10,000 test)
Model Structure	Spiking Convolutional Neural Network (<code>cnn_mnist_snn</code>)
Loss Function	Rate-coded Cross-Entropy Loss (<code>ce_rate_loss</code>)
Optimizer	SGD
Learning Rate	0.10
Momentum	0.9
Weight Decay	0.0001
Batch Size	128
SNN Encoding	Constant encoding, 10 time steps
SNN Decay (β)	0.95
SNN Threshold	1.0 (fixed)
Training Steps	500
Training Algorithm	DSGD
Honest Nodes (N)	10
Byzantine Nodes (f)	$f \in \{0, 1, 2, 3, 4, 5\}$
Data Heterogeneity (γ)	$\gamma \in \{1.0, 0.66, 0.33, 0.0\}$ (from IID to extreme Non-IID)
Pre-aggregators	NNM, ARC
Aggregators Swept	Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean
Attacks Evaluated	Optimal ALIE, Sign Flipping, Optimal Inner Product Manipulation
Training Seeds	5 seeds (42 to 46)
Data Dist. Seed	42

2 Best Overall Performance (Worst-Case Across Attacks)

The heatmaps in this section display the worst-case test accuracy at convergence across all attacks achieved by selecting the best robust aggregator for each cell.

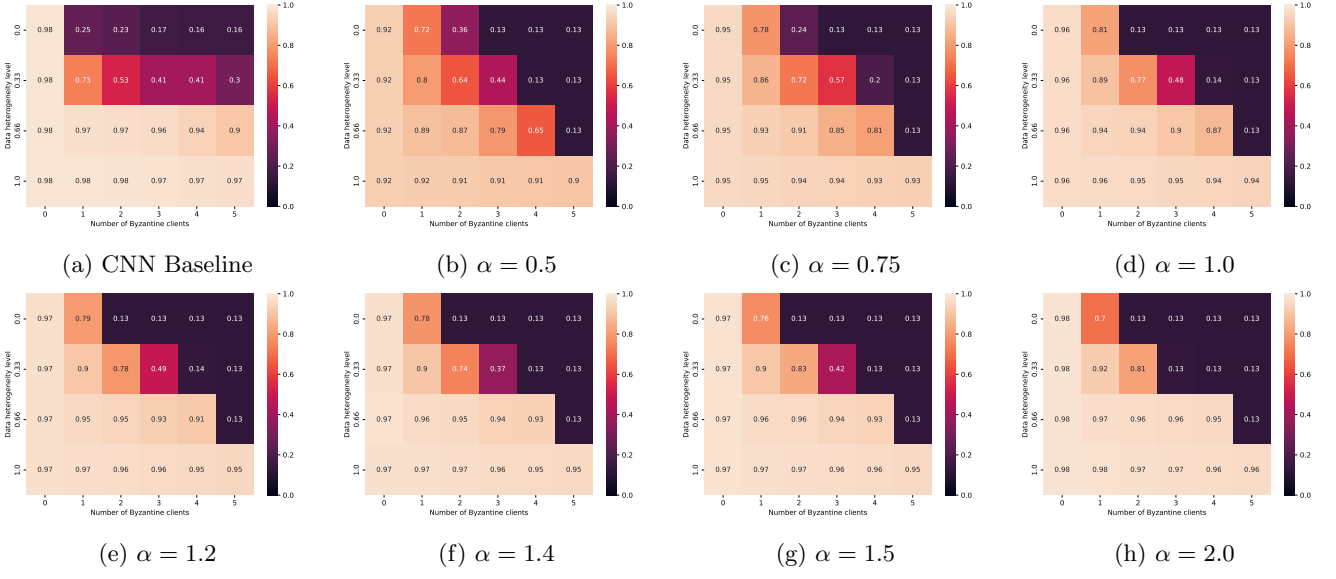


Figure 1: Best overall test accuracy heatmaps (worst-case across attacks) for different α stiffness parameter values.

3 Best Aggregator under Specific Attacks

This section presents heatmaps showing the best aggregator performance under each specific attack across all alphas.

3.1 Optimal A Little Is Enough (ALIE) Attack

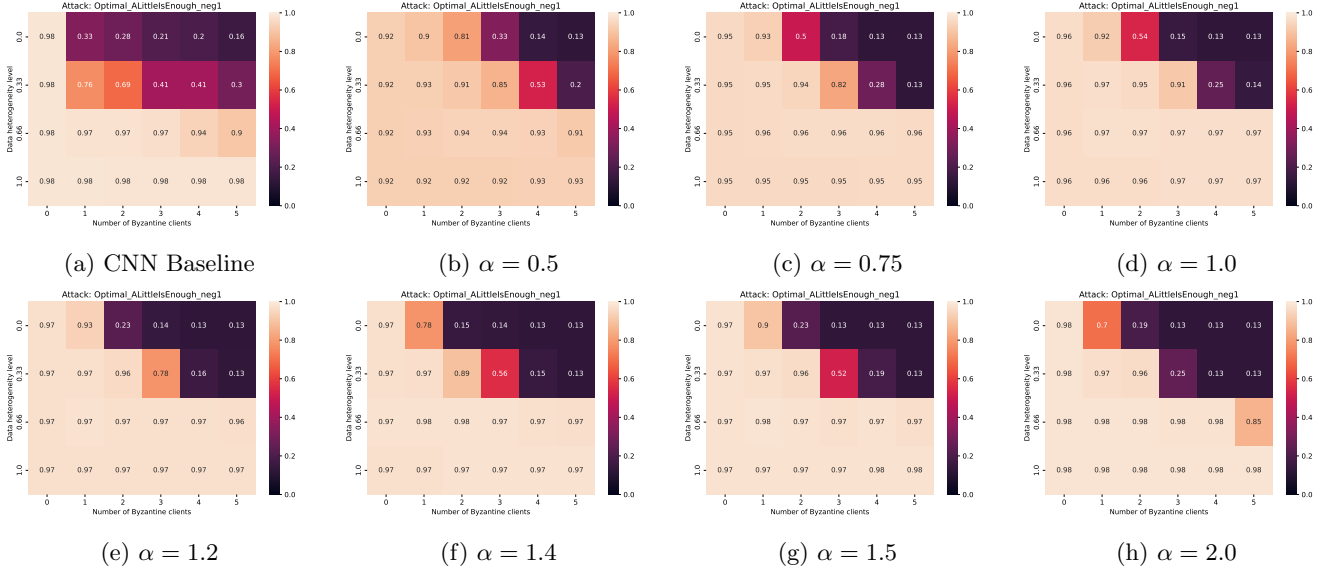


Figure 2: Best test accuracy heatmaps under the Optimal ALIE attack for different α parameter values.

3.2 Sign Flipping Attack

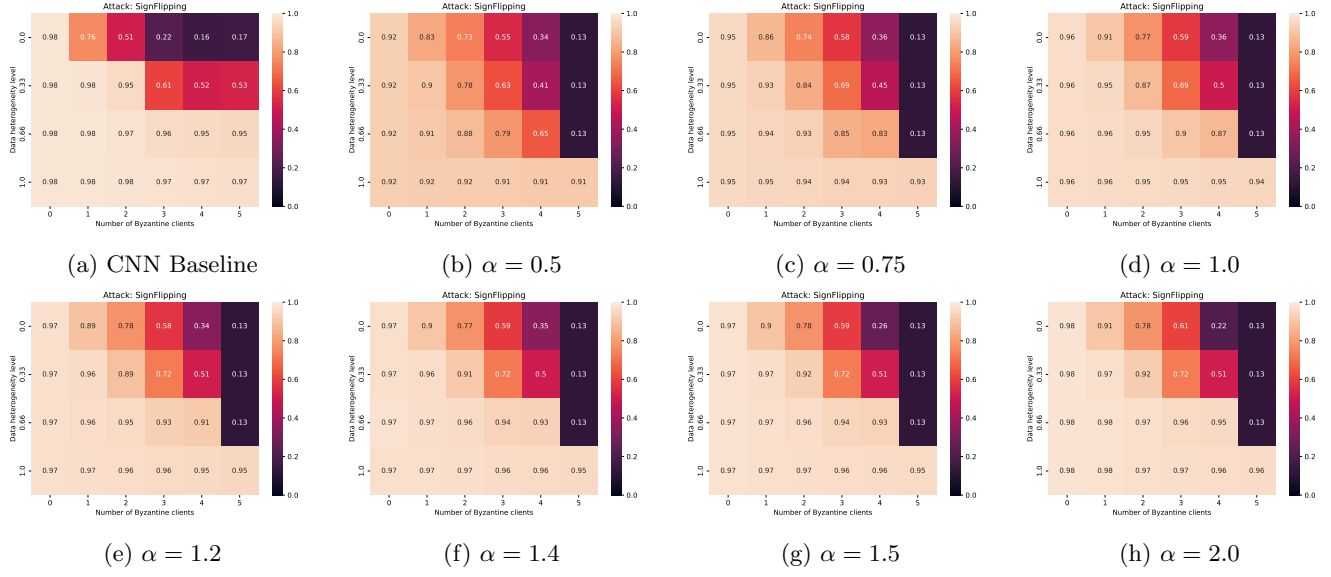


Figure 3: Best test accuracy heatmaps under the Sign Flipping attack for different α parameter values.

3.3 Optimal Inner Product Manipulation Attack

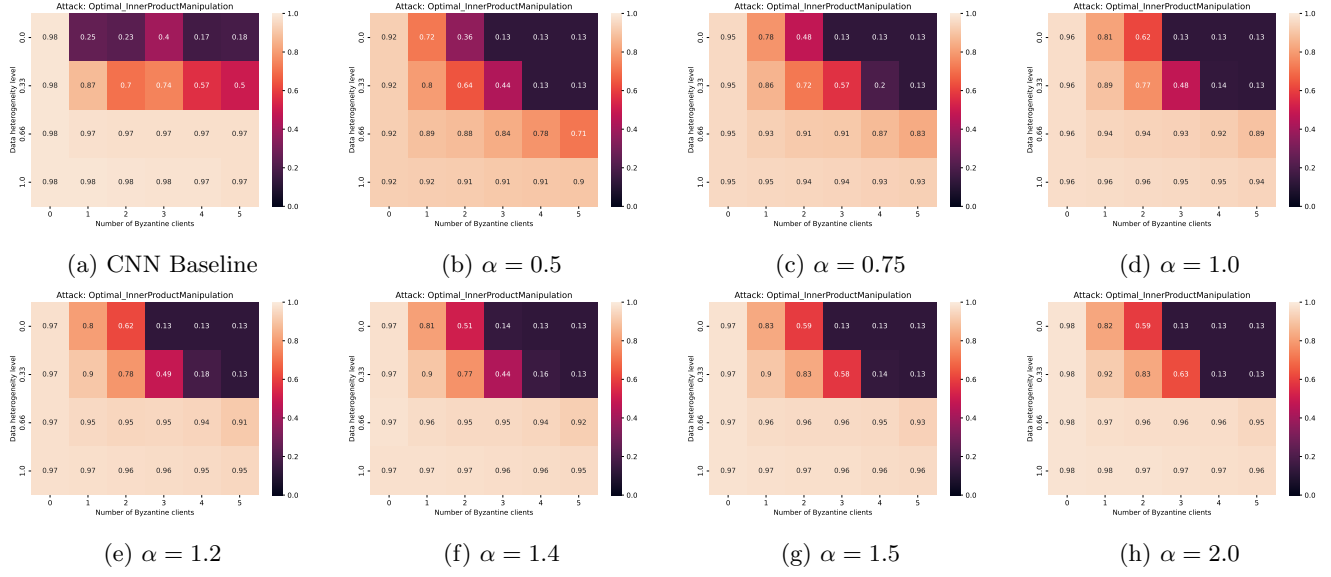


Figure 4: Best test accuracy heatmaps under the Optimal Inner Product Manipulation attack for different α parameter values.

4 Robustness of Fixed Aggregators across all Attacks

In this section, we analyze the performance of each specific aggregator (Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean) across all Byzantine attack settings for different stiffness α values.

4.1 Centered Clipping (CC)

4.1.1 CC under Worst-Case across all Attacks

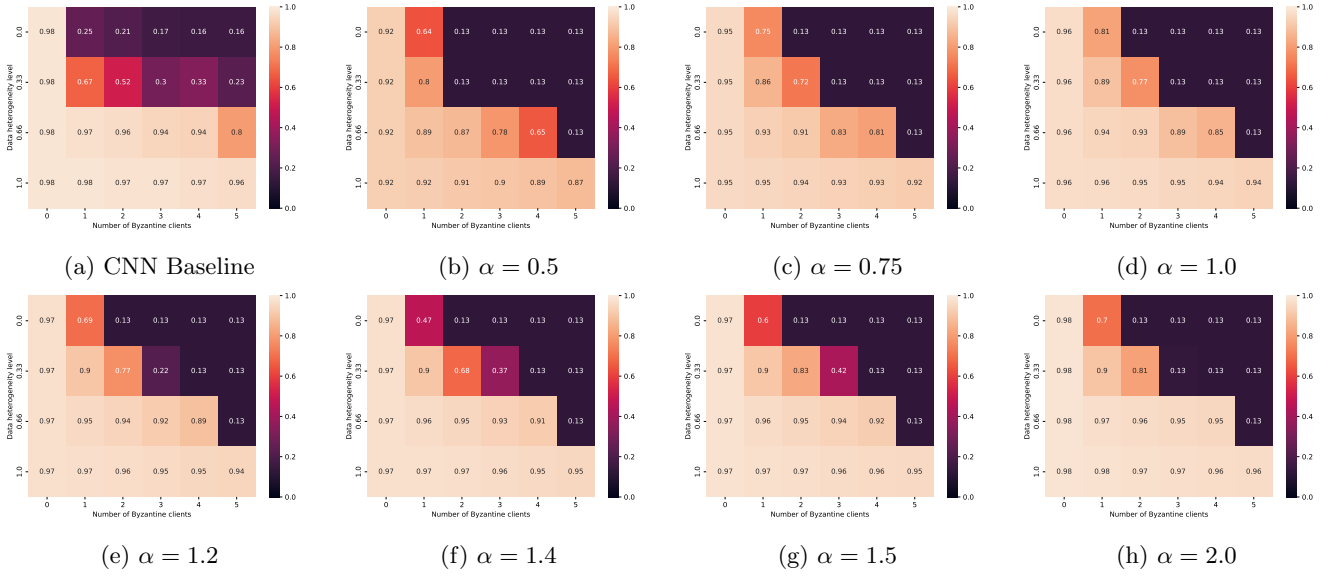


Figure 5: Centered Clipping performance under the worst-case across all attacks across α values.

4.1.2 CC under Optimal ALIE Attack

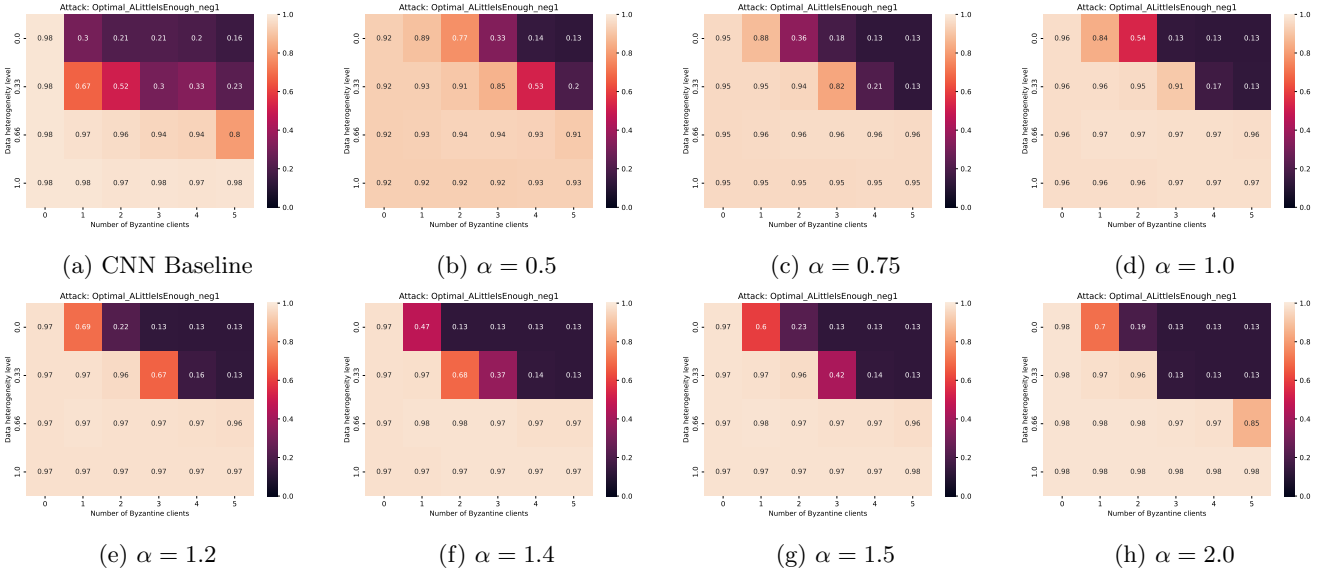


Figure 6: Centered Clipping performance under Optimal ALIE across α values.

4.1.3 CC under Sign Flipping Attack

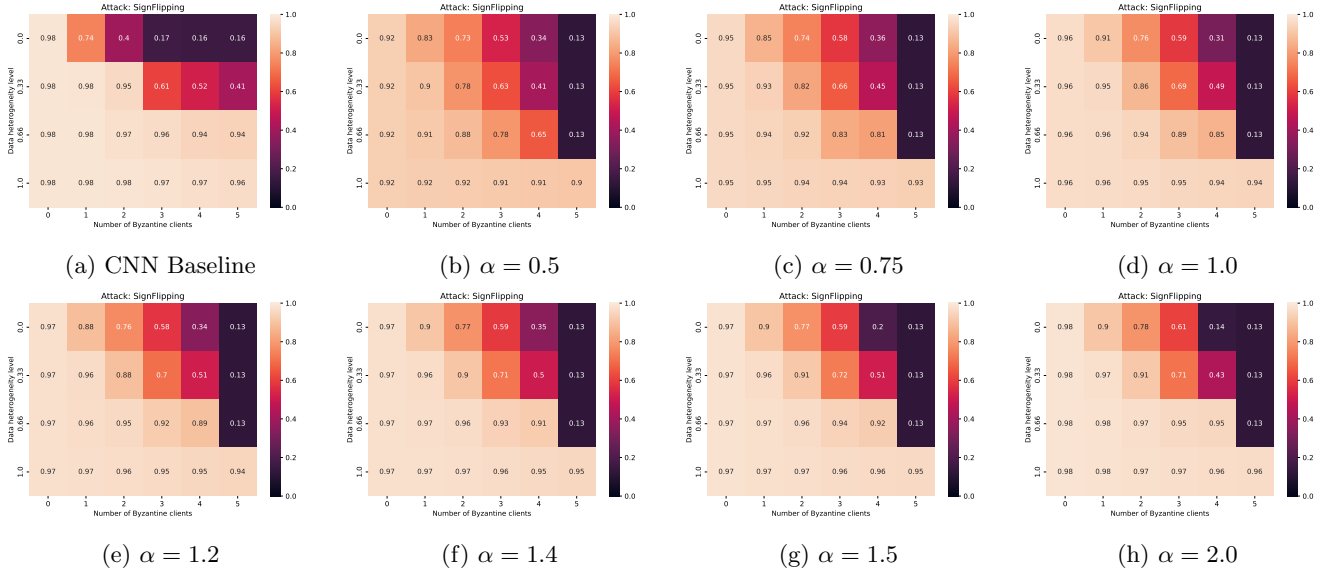


Figure 7: Centered Clipping performance under Sign Flipping across α values.

4.1.4 CC under Optimal Inner Product Manipulation Attack

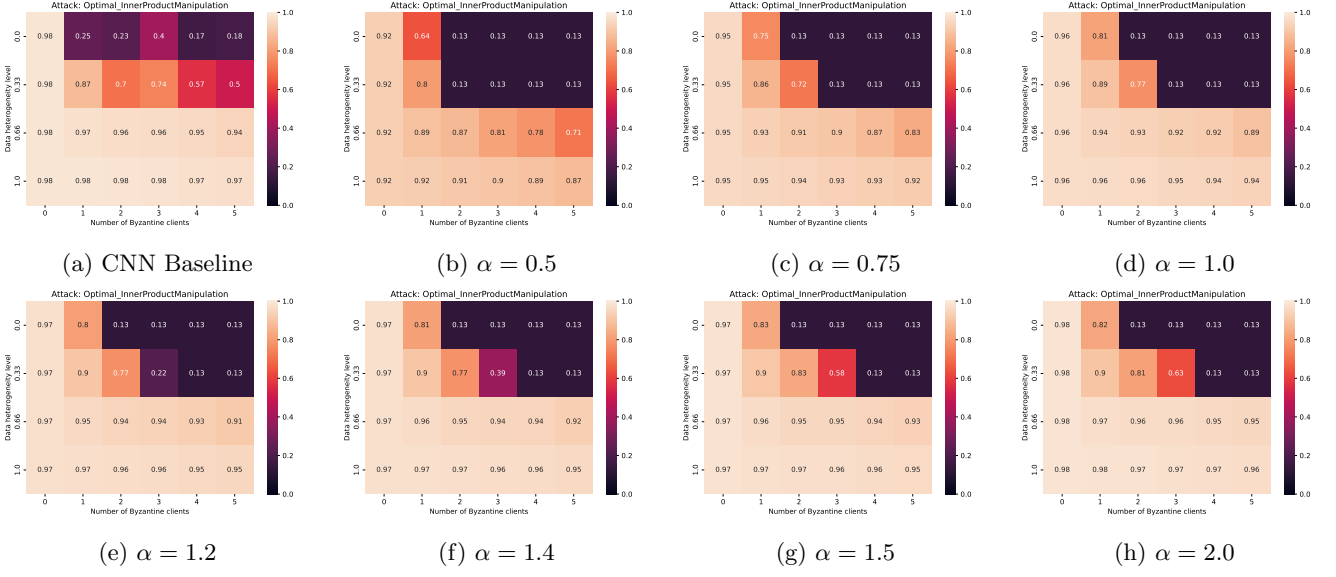


Figure 8: Centered Clipping performance under Optimal Inner Product Manipulation across α values.

4.2 Geometric Median (GM)

4.2.1 GM under Worst-Case across all Attacks

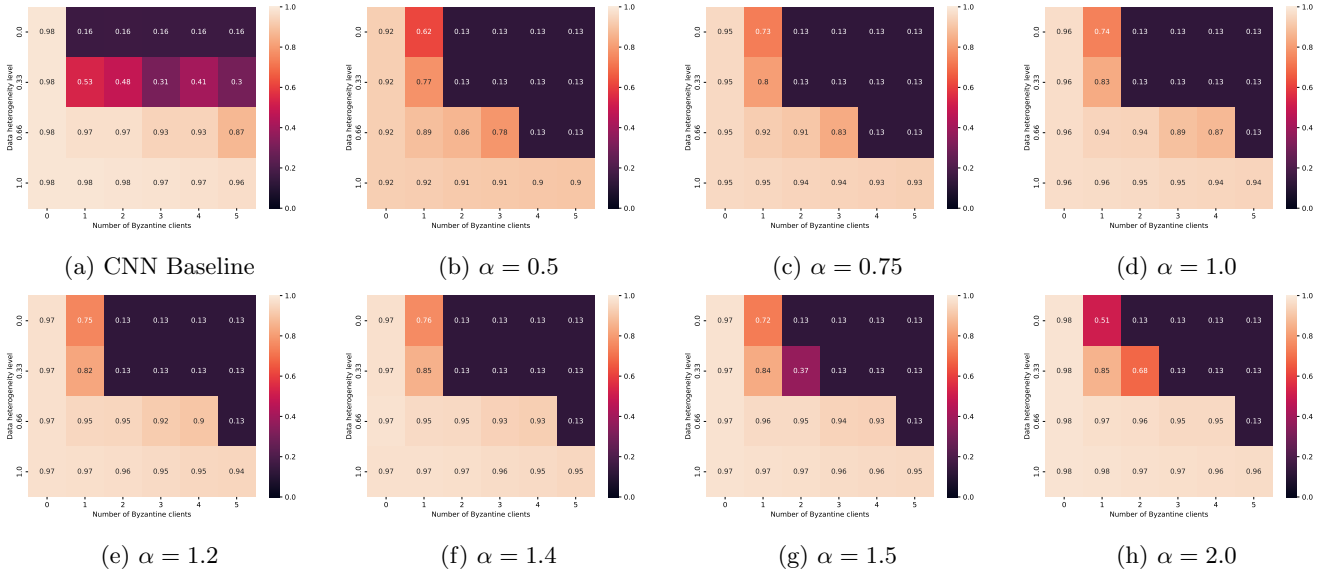


Figure 9: Geometric Median performance under the worst-case across all attacks across α values.

4.2.2 GM under Optimal ALIE Attack

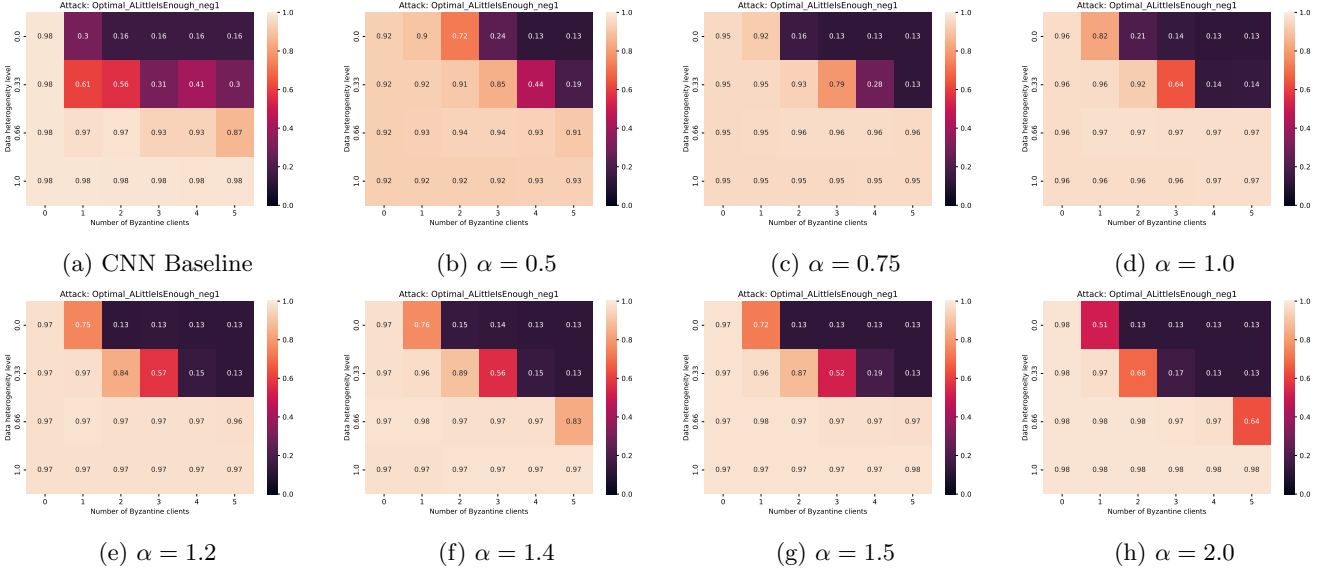


Figure 10: Geometric Median performance under Optimal ALIE across α values.

4.2.3 GM under Sign Flipping Attack

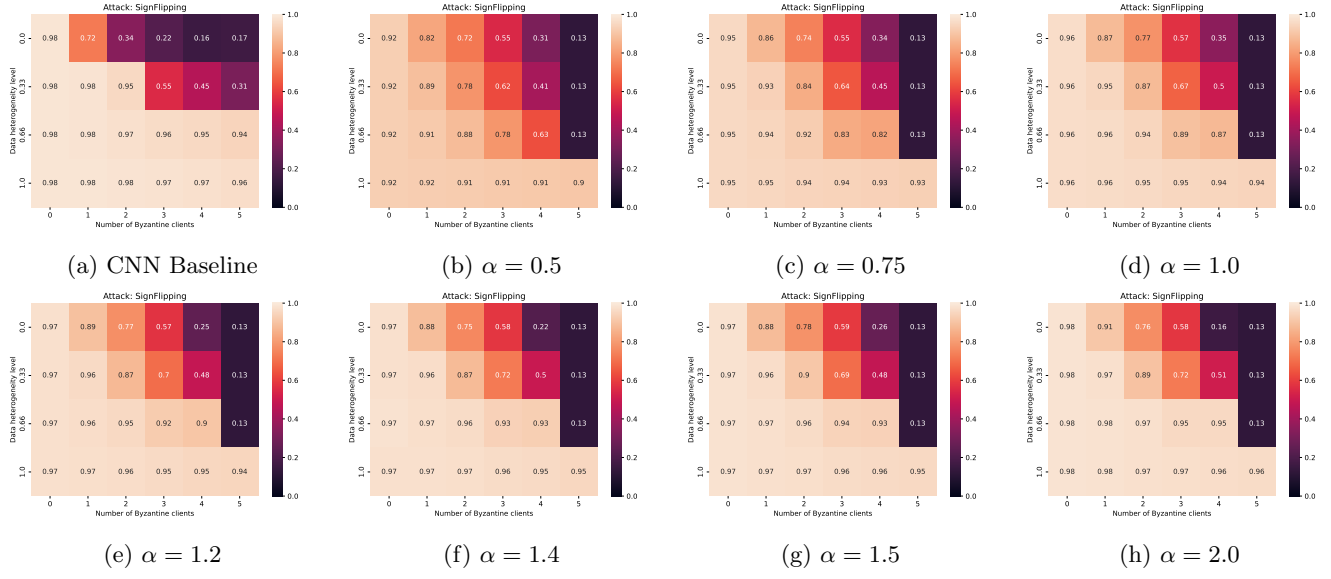


Figure 11: Geometric Median performance under Sign Flipping across α values.

4.2.4 GM under Optimal Inner Product Manipulation Attack

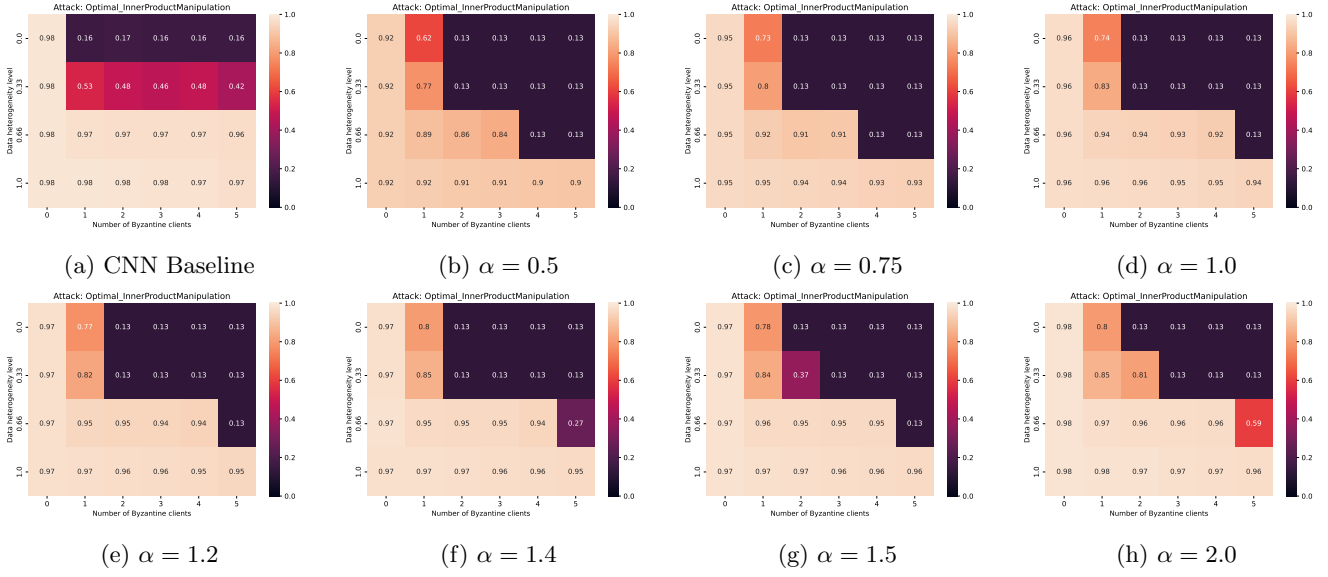


Figure 12: Geometric Median performance under Optimal Inner Product Manipulation across α values.

4.3 Multi-Krum (MK)

4.3.1 MK under Worst-Case across all Attacks

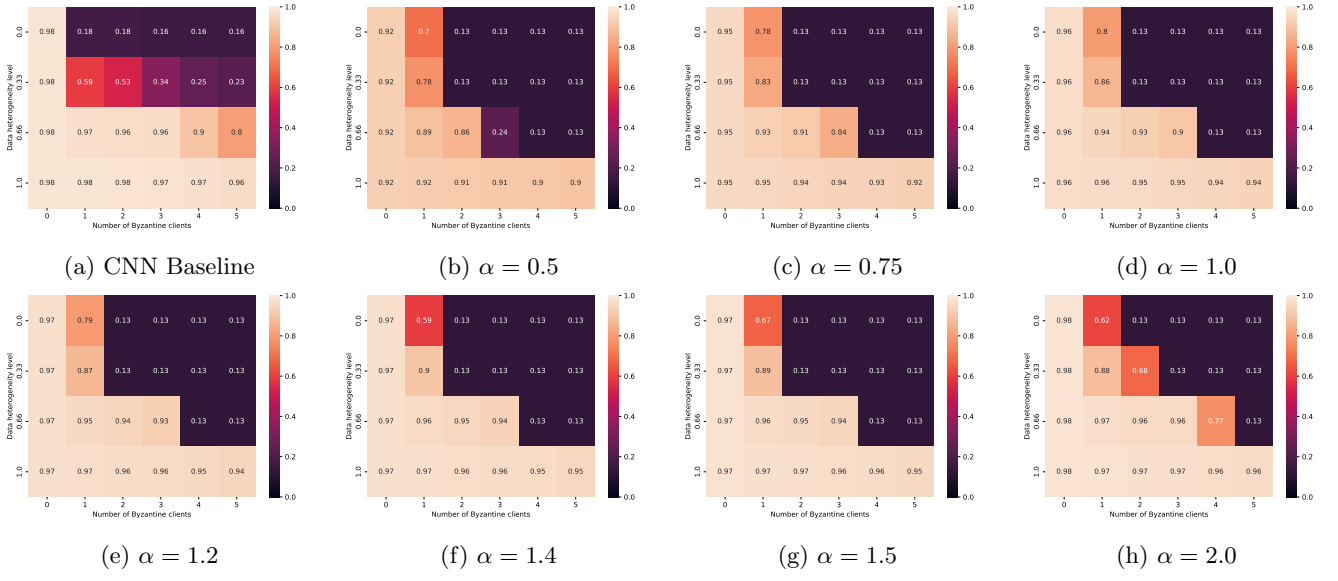


Figure 13: Multi-Krum performance under the worst-case across all attacks across α values.

4.3.2 MK under Optimal ALIE Attack

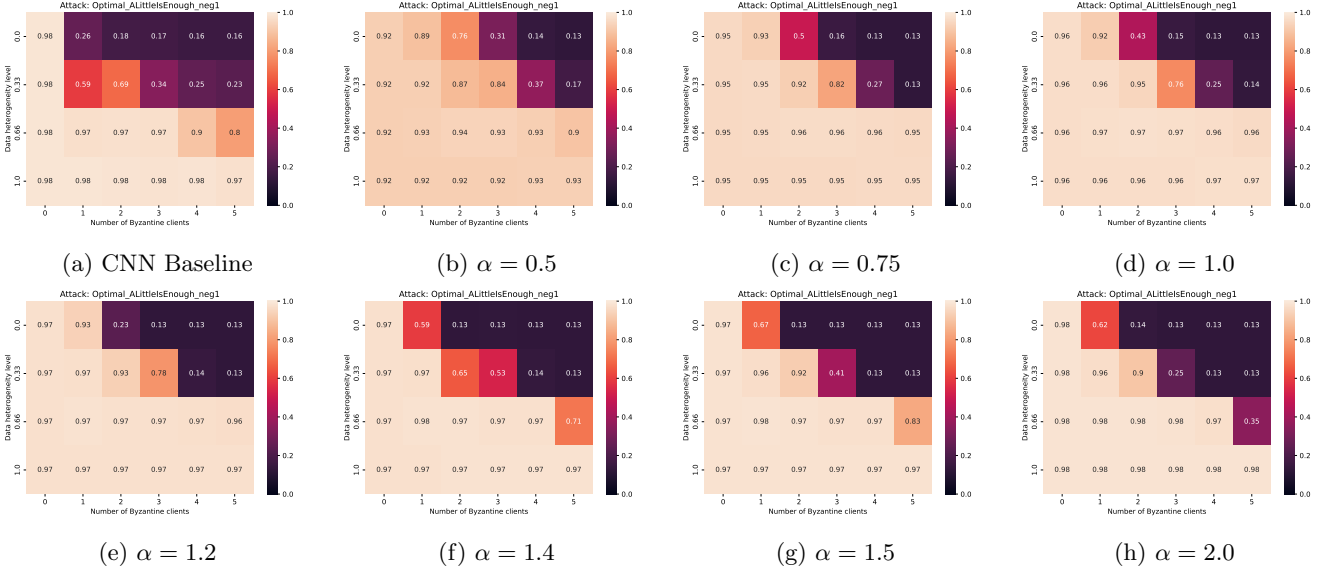


Figure 14: Multi-Krum performance under Optimal ALIE across α values.

4.3.3 MK under Sign Flipping Attack

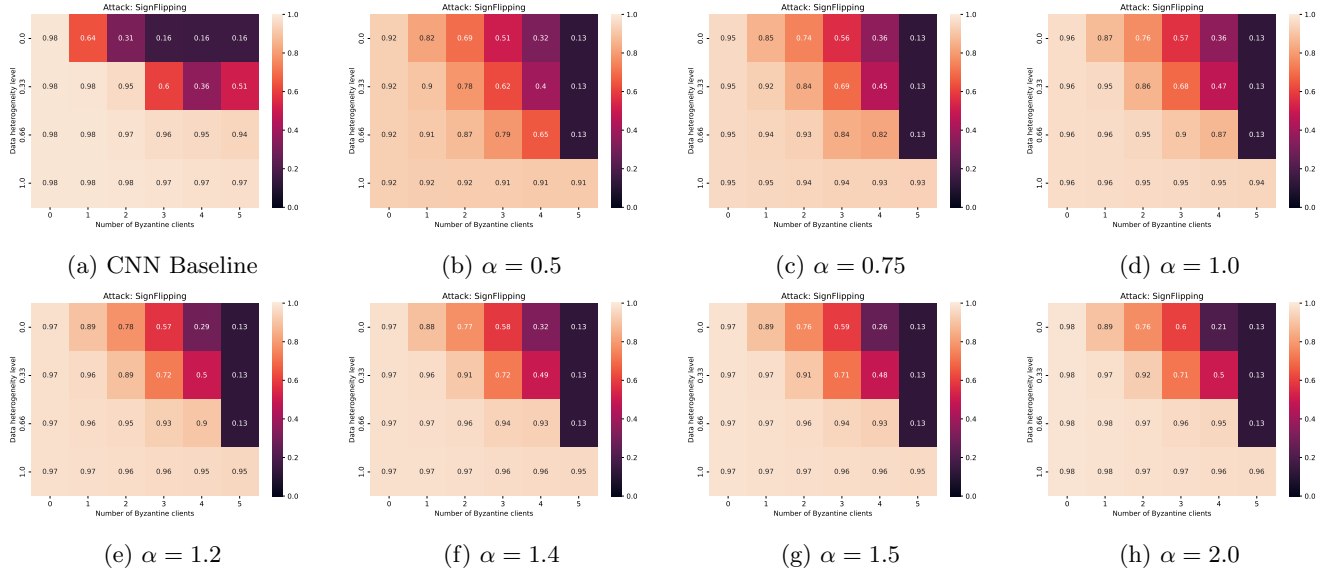


Figure 15: Multi-Krum performance under Sign Flipping across α values.

4.3.4 MK under Optimal Inner Product Manipulation Attack

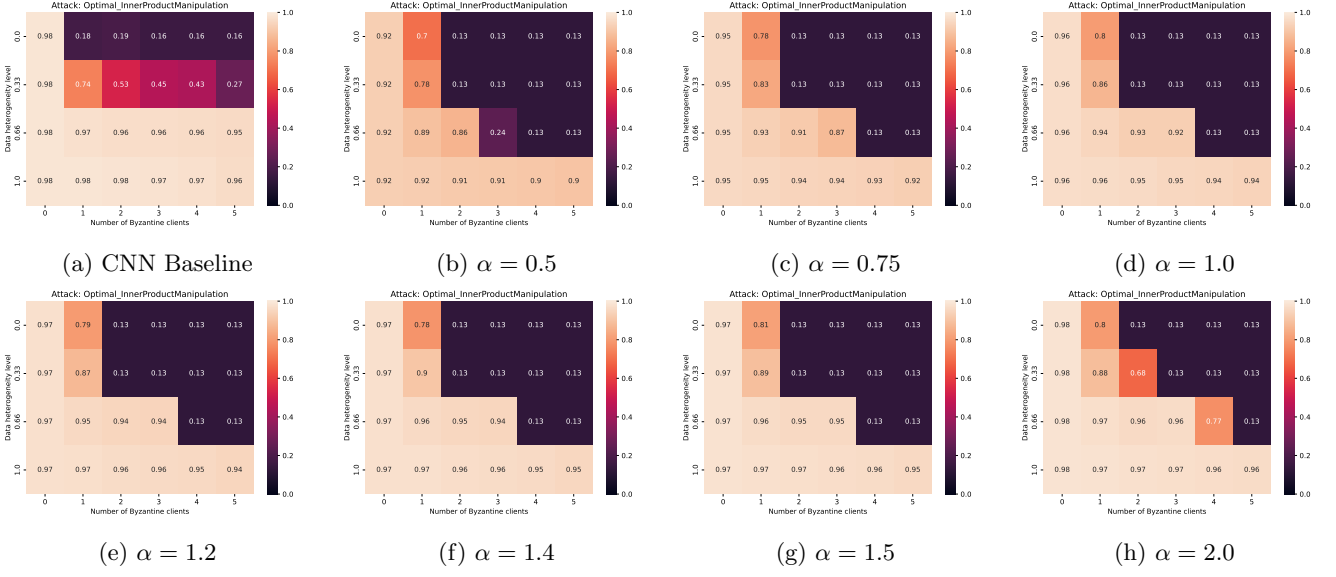


Figure 16: Multi-Krum performance under Optimal Inner Product Manipulation across α values.

4.4 Trimmed Mean (TM)

4.4.1 TM under Worst-Case across all Attacks

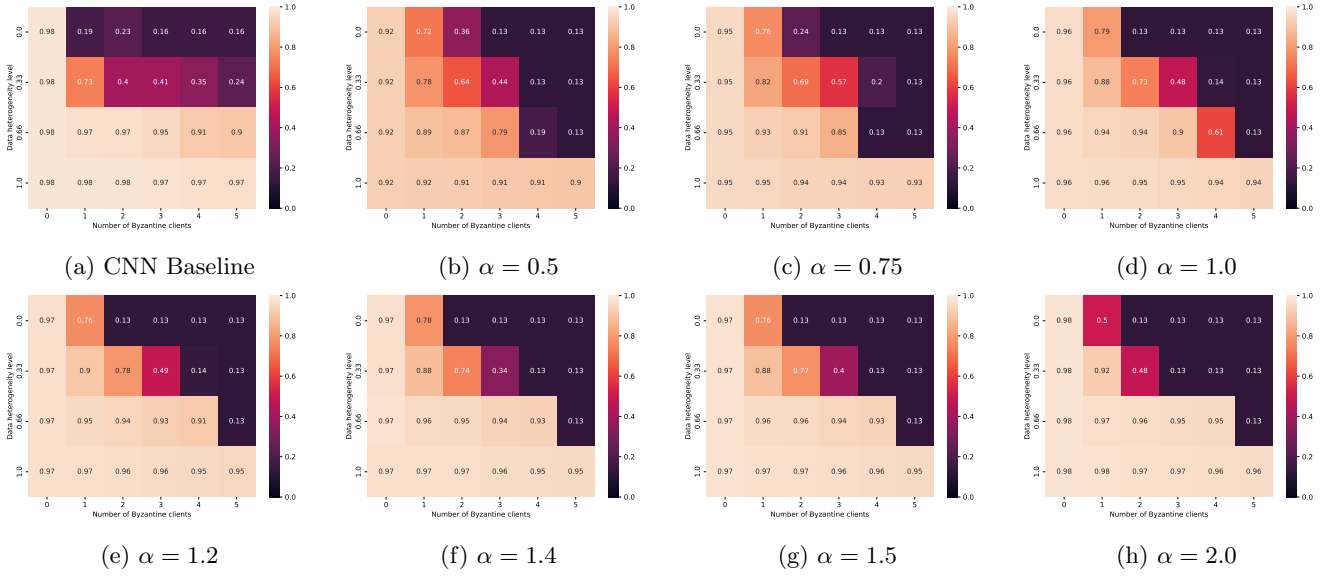


Figure 17: Trimmed Mean performance under the worst-case across all attacks across α values.

4.4.2 TM under Optimal ALIE Attack

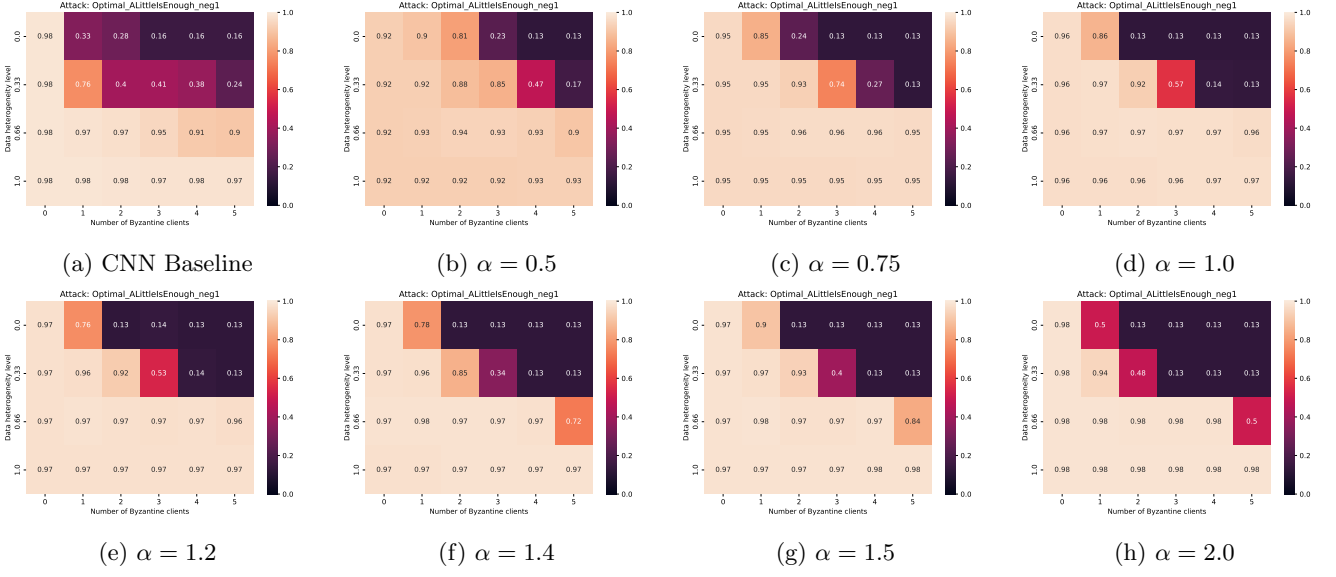


Figure 18: Truncated Mean performance under Optimal ALIE across α values.

4.4.3 TM under Sign Flipping Attack

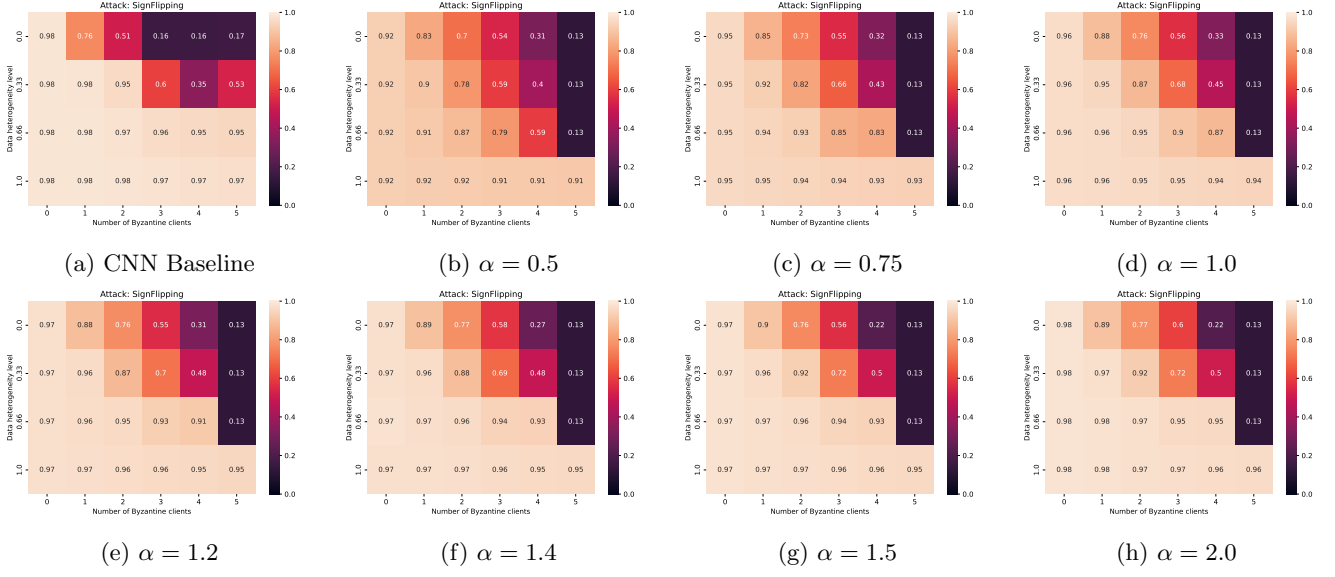


Figure 19: Truncated Mean performance under Sign Flipping across α values.

4.4.4 TM under Optimal Inner Product Manipulation Attack

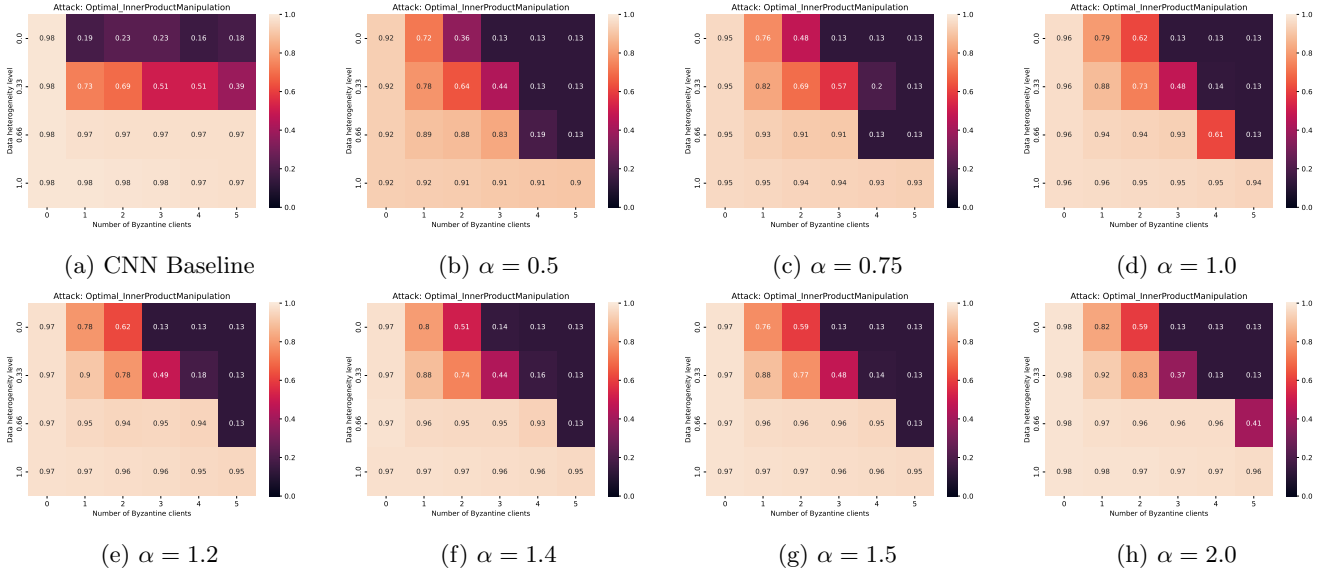


Figure 20: Truncated Mean performance under Optimal Inner Product Manipulation across α values.